bases (B_1 *and* B_2) is called inter-base resistance and is very high (5 to 10 kΩ), when emitter is in open condition.

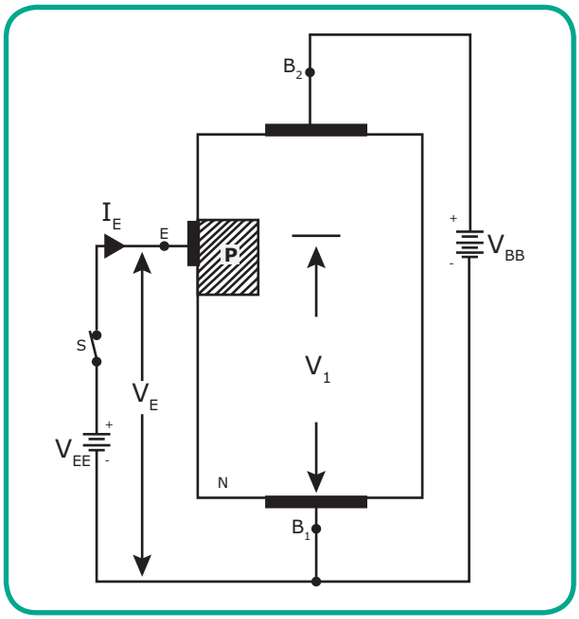


Figure 6.8(b) UJT Biasing

Working Principle: Figure 6.8(b) shows the biasing arrangement of UJT. The base B_2 is kept at positive potential with respect to B_1 using the battery terminal V_{BB}. The *PN* junction is always forward biased by making emitter (*E*) positive with respect to base B_1 using the battery terminal V_{EE}.

When No Voltage is Applied at the Emitter: The emitter is open (switch *S* is open). When a voltage V_{BB} is applied between the two bases, a voltage gradient is established along the N-type silicon bar. The voltage drop (V_1) between emitter and base B_1, reverse bias the *PN* junction and hence no emitter current flows (i.e. UJT is in *OFF* state). However, a small leakage current flows from base B_2 to emitter *E*.

When a Positive Voltage is Applied at the Emitter: When a positive voltage is applied to the emitter. (Switch *S* is closed). The *PN* junction is also reverse biased. If the emitter voltage is linearly increased, which reduces the amount of reverse bias of the *PN* junction. When the applied voltage is increased and greater than V_1, the *PN* junction will be forward biased. Now, the holes are injected from P-region into N-region. The holes are repelled by the terminal B_2 and are attracted by the terminal B_1. Now, the accumulation of holes in *E* to B_1region reduces the resistance in this section. Hence, if the emitter current I_E and the voltage V_E increases, the UJT becomes in the *ON* state.

When a Negative Voltage is Applied at the Emitter: If a negative voltage is applied to the emitter (*E*), the *PN* junction is reverse biased, and the emitter current is very low. The device is now in the *OFF* state.

Application

1. UJT is used as relaxation oscillator.
2. It is widely used as triggering device for SCR and TRIAC.
3. It is used in phase control circuits.
4. UJTs can also be used to measure magnetic flux.
5. It is used in switching circuits
6. It is used as sawtooth generator.
7. It is used in tuning circuits (TV).

6.9. FIELD EFFECT TRANSISTOR (FET)

In the previous Chapters, we have discussed about the Bipolar Junction Transistor (BJT). They are controlled by both electrons and holes and called current operated devices. BJT has two main disadvantages; first, it has low input impedance because of forward biased emitter junction. Secondly, it has considerable noise level.

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